

FORTRESS INDUSTRIAL PLANT - ASSET HEALTH REPORT

DOCUMENT ID: REP-2026-P101-V4

EQUIPMENT TAG: PUMP P-101 (PRIMARY FEED WATER BOOSTER)

LOCATION: SECTOR 4, TURBINE AUXILIARY BAY

DATE OF INSPECTION: AUGUST 14, 2026

LEAD INSPECTOR: MARCUS VANCE (CERTIFIED ISO 18436-2 CATEGORY III)

1. EXECUTIVE SUMMARY

A comprehensive condition monitoring and vibration analysis was conducted on Pump P-101 following reports of abnormal acoustics.

The main issue identified in Pump P-101 is severe mechanical vibration and advanced outer race degradation on the drive-end bearing.

2. OPERATIONAL PARAMETERS RECORDED

- Operating Speed: 2,980 RPM
- Flow Rate: 420 m3/hr (Nominal: 450 m3/hr)
- Suction Pressure: 3.2 bar
- Discharge Pressure: 18.6 bar
- Motor Drive Current: 84.5 Amperes

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3. VIBRATION ANALYSIS & MEASUREMENTS

Overall vibration levels were measured using triaxial accelerometers mounted on non-drive-end (NDE) and drive-end (DE) bearings.

- Measured Overall Vibration Level (Drive-End Radial): 7.2 mm/s RMS (ISO 10816-3 Zone D: Critical/Unacceptable)
- Baseline Acceptable Vibration: 1.8 mm/s RMS
- Peak Acceleration: 4.8 g pk-pk in the 1.2 kHz - 3.5 kHz demodulated frequency spectrum.
- High amplitude peaks observed at 118.4 Hz matching the Outer Race Ball Pass Frequency (BPFO).

4. THERMOGRAPHIC FINDINGS

- Drive-end bearing housing skin temperature: 88.4 deg C (Exceeds maximum allowable limit of 70 deg C).
- Lubricant oil sample: High iron particulate concentration (142 ppm), indicating active metal-to-metal spalling.

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5. ROOT CAUSE ASSESSMENT

The excessive vibration and thermal runaway are caused by lubrication starvation and contamination, which resulted in micro-s

6. RECOMMENDED MAINTENANCE ACTIONS

The following immediate maintenance actions are mandated:

- Action 1 (Critical): Perform an immediate scheduled shutdown of Pump P-101 within 14 days.
- Action 2: Replace drive-end bearing (SKF 22218 EK) and inspect shaft sleeve for fretting corrosion.
- Action 3: Flush the oil reservoir and replenish with ISO VG 46 synthetic lubricating oil.
- Action 4: Verify dynamic alignment between motor and pump coupling (tolerance < 0.05 mm radial runout).
- Action 5: Re-commission and perform post-maintenance vibration verification.

REPORT APPROVED BY:

H. Thorne, Head of Plant Reliability Engineering